

Practice Test I (Version 2)

Name
Date
Course

- I) Write your name, the date, and the course title (i.e. Precalculus).
- II) Show all work.
- III) Draw a rectangle around your final answer for each problem.
- IV) Calculators are not allowed for this test.
- V) Each problem is worth four points for a total of 100 points.

① Simplify the polynomial below, write in descending order of powers, and identify the degree.

$$-3x + 2x - 14x^3 + 2x^4 - x^3 + 6$$

② Subtract

$$(4x^3 - 8x^2 + 15x - 12) - (x^3 - x^2 + 10)$$

③ What is the maximum number of turning points of the graph of the function below?

$$x^3 - 3x^2 + x - 8$$

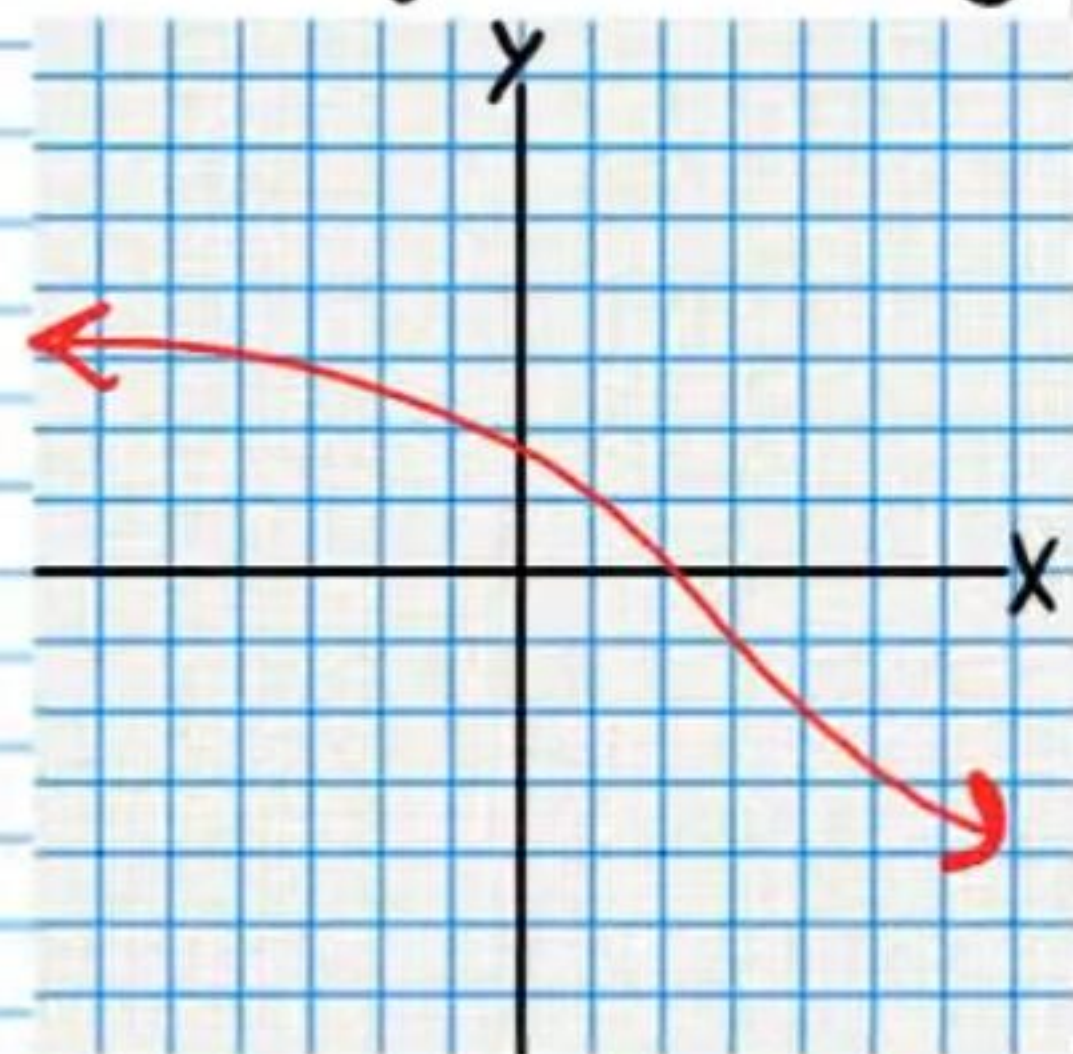
④ Indicate the end behavior of the polynomial below.

$$f(x) = -2x^4 - 4x + 1$$

⑤ The graph of a function is written below. Write the intervals where the function is increasing or decreasing.

Increasing:

Decreasing:



⑥ Use the Rational Zeros Theorem to find all of the potential rational zeros of the function below.

$$f(x) = 4x^7 - 3x^5 + x^3 - 2x + 5$$

⑦ Write the zeros of the polynomial below, the multiplicity of each zero, and whether or not the corresponding graph crosses or simply touches the x-axis at each zero.

$$f(x) = -6(x+7)^5(x+3)^2(x+6)$$

⑧ Explain why you cannot use synthetic division to perform the following operation.

$$(10x^4 - 3x^3 + 2) \div (2x - 1)$$

⑨ Divide using synthetic division.

$$(1 + x + x^3 + x^4) \div (x + 1)$$

⑩ Write a polynomial function in standard form with real coefficients that has the following zeros, leading coefficient (LC), and degree (D).

$$2, -2, 3, 1 \quad LC = -1 \quad D = 4$$

⑪ Find the zeros of the following function using the grouping method. Find all of them, whether rational, irrational, or imaginary.

$$f(x) = 3x^4 - 4x^3 - 81x + 108$$

Solve. Write the solutions graphically and in interval notation.

⑫ $x^2 \leq 9$

⑬ $x^2 + 2 < 4x$

$$(14) \frac{x+4}{x-1} \geq \frac{x-2}{x-3}$$

$$(15) \frac{3x^3 - 21x^2 + 30x}{x^4 + x^2 - 2} > 0$$

Graph the following functions. Draw asymptotes as dotted lines and write their equations. Also find intercepts.

$$(16) f(x) = \frac{2x-4}{x+1}$$

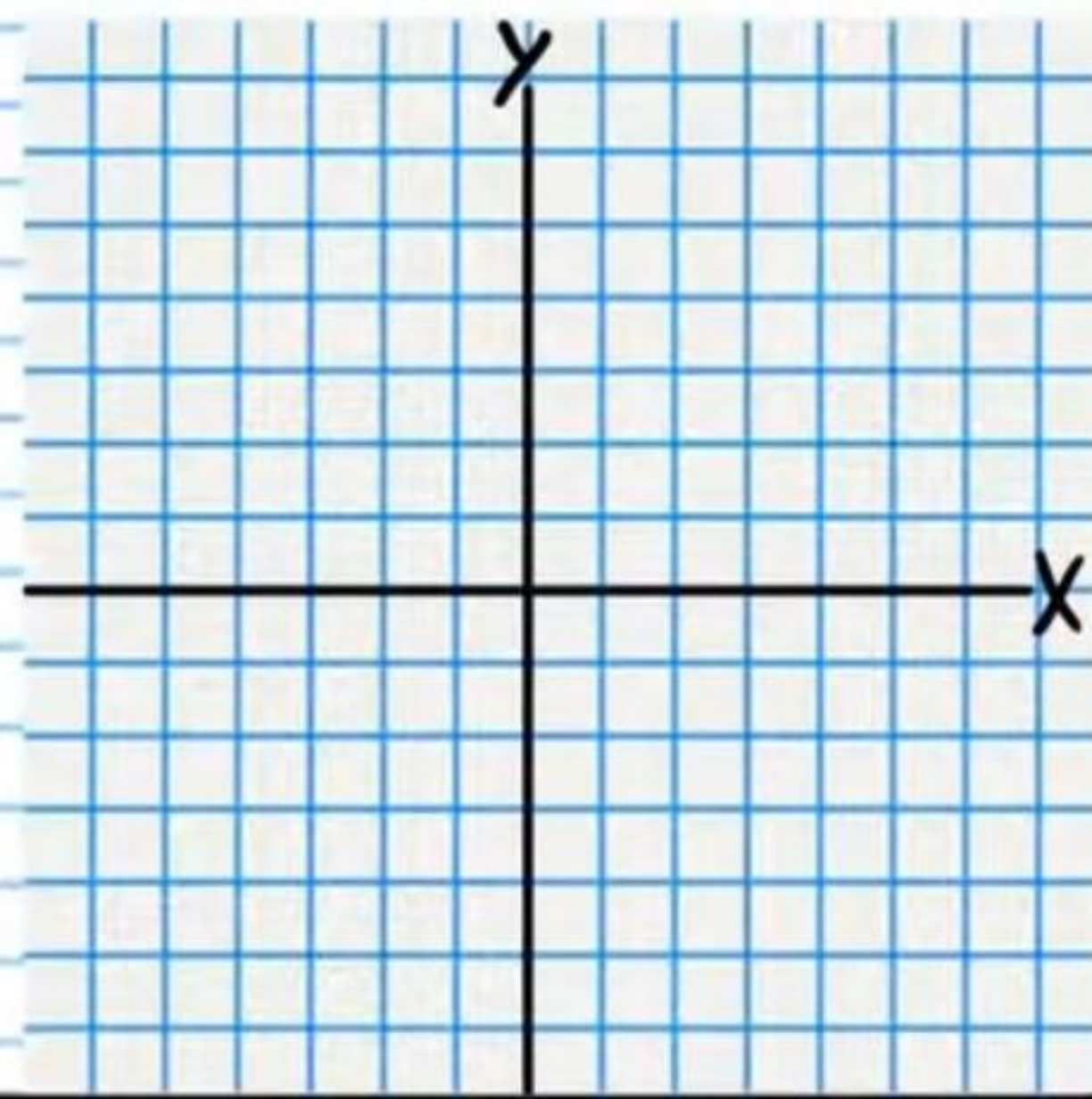
Horizontal Asymptote Equation

Vertical Asymptote Equation(s)

Sign Chart

Y-Intercept

X-Intercept



⑪ $f(x) = \frac{6x-18}{x^2-2x-3}$

Hole

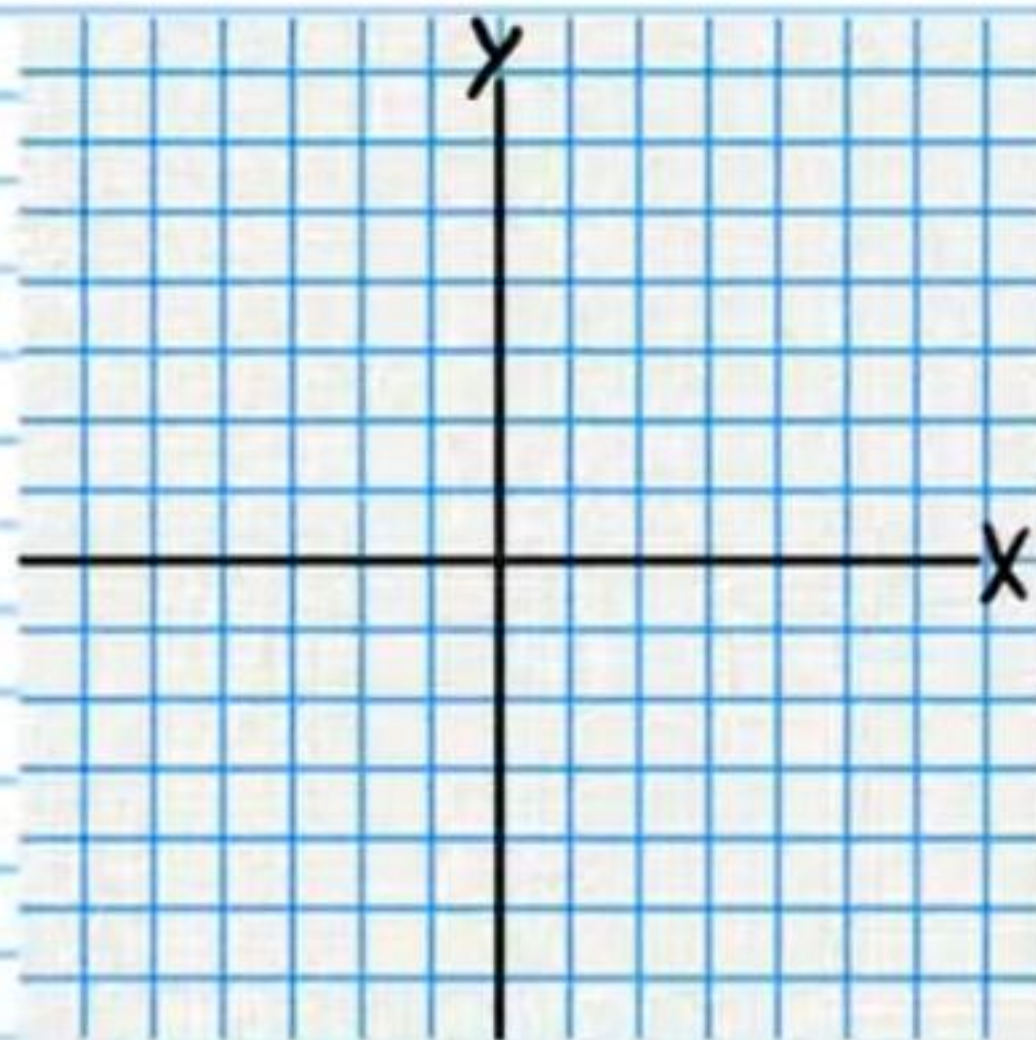
Vertical Asymptote Equations

Horizontal Asymptote Equations

Sign Chart

Y-Intercept

X-Intercept



⑫ $f(x) = -\frac{2}{x^2+6x+8}$

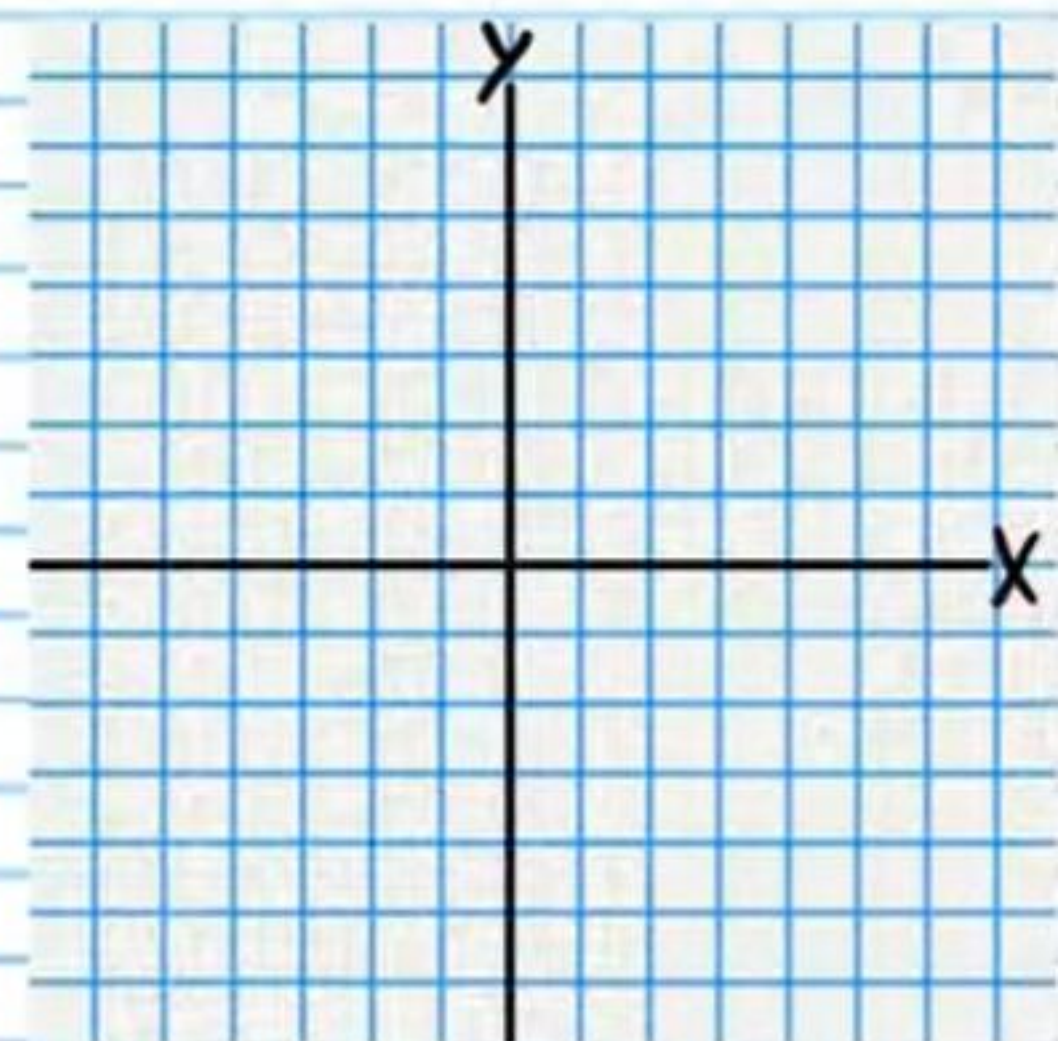
Vertical Asymptote Equations

Horizontal Asymptote Equations

Sign Chart

Y-Intercept

X-Intercept



①9 Divide using synthetic division.

$$(x^4 - 1) \div (-1 + x)$$

②0 Divide.

$$(2x + 2x^4 - 1) \div (-x + 2)$$

②1 Find the roots of the polynomial function below.
Write all of them, whether rational, irrational, or imaginary.

$$f(x) = x^4 + 3x^3 - 6x - 4$$

②② Find the zeros of the function below. Write all of them, whether rational, irrational, or imaginary.

$$f(x) = 9x^3 - 3x^2 - 7x + 5$$

②③ Find the x-intercepts of the graph of the function below and determine if the graph crosses or touches the x-axis at each intercept. Write the function in factored form.

$$f(x) = x^4 - 8x^3 + 18x^2 - 27$$

②4 Find the solutions of the equation below.

$$2x^5 - x^4 - 16x^3 + 8x^2 + 32x - 16 = 0$$

②5 For the function below, find the information requested.
It is not necessary to graph the function.

$$f(x) = \frac{x^3 + 4x^2 + 2x + 8}{x + 2}$$

Y-Intercept

X-Intercept

Vertical Asymptote Equation(s)

Asymptote Curve

Homework Answers

① $2x^4 - 15x^3 - x + 6, D=4$

② $3x^3 - 7x^2 + 15x - 22$

③ 2

④ As $x \rightarrow \infty, f(x) \rightarrow -\infty$
As $x \rightarrow -\infty, f(x) \rightarrow -\infty$

⑤ Increasing: x
Decreasing: $[-7, 7]$

⑥ $\pm 1, \pm 5, \pm \frac{1}{2}, \pm \frac{1}{4}, \pm \frac{5}{2}, \pm \frac{5}{4}$

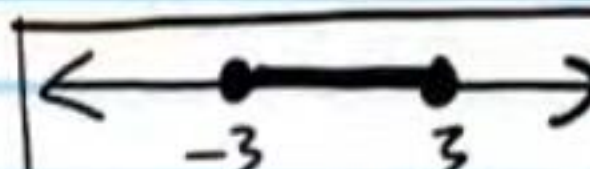
⑦ $-7 \Rightarrow$ Multiplicity: 5 $\Rightarrow$ crosses x-axis
 $-3 \Rightarrow$ Multiplicity: 2 $\Rightarrow$ Touches x-axis
 $-6 \Rightarrow$ Multiplicity: 1 $\Rightarrow$ crosses x-axis

⑧ The divisor is not a binomial of the form $x-a$ or $x+a$

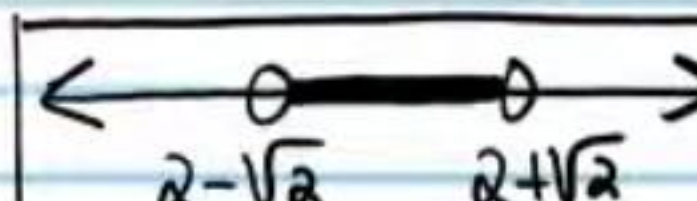
⑨ $x^3 + 1$

⑩ $-x^4 + 4x^3 + x^2 - 16x + 12$

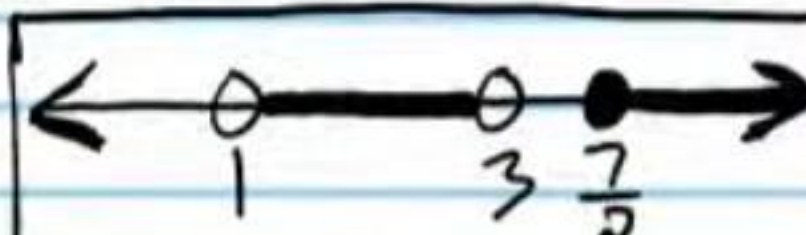
⑪ $3, \frac{4}{3}, \frac{-3+3i\sqrt{3}}{2}, \frac{-3-3i\sqrt{3}}{2}$

⑫ 

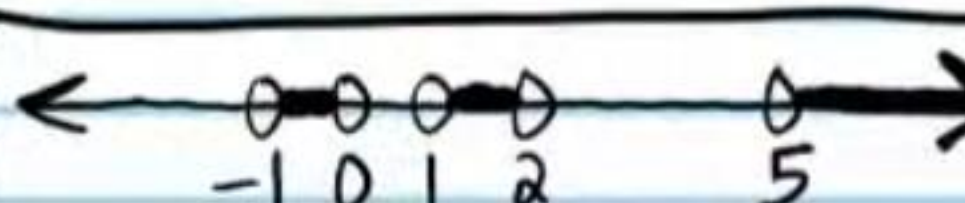
$[-3, 3]$

⑬ 

$(2-\sqrt{2}, 2+\sqrt{2})$

⑭ 

$(1, 3) \cup [\frac{7}{2}, \infty)$

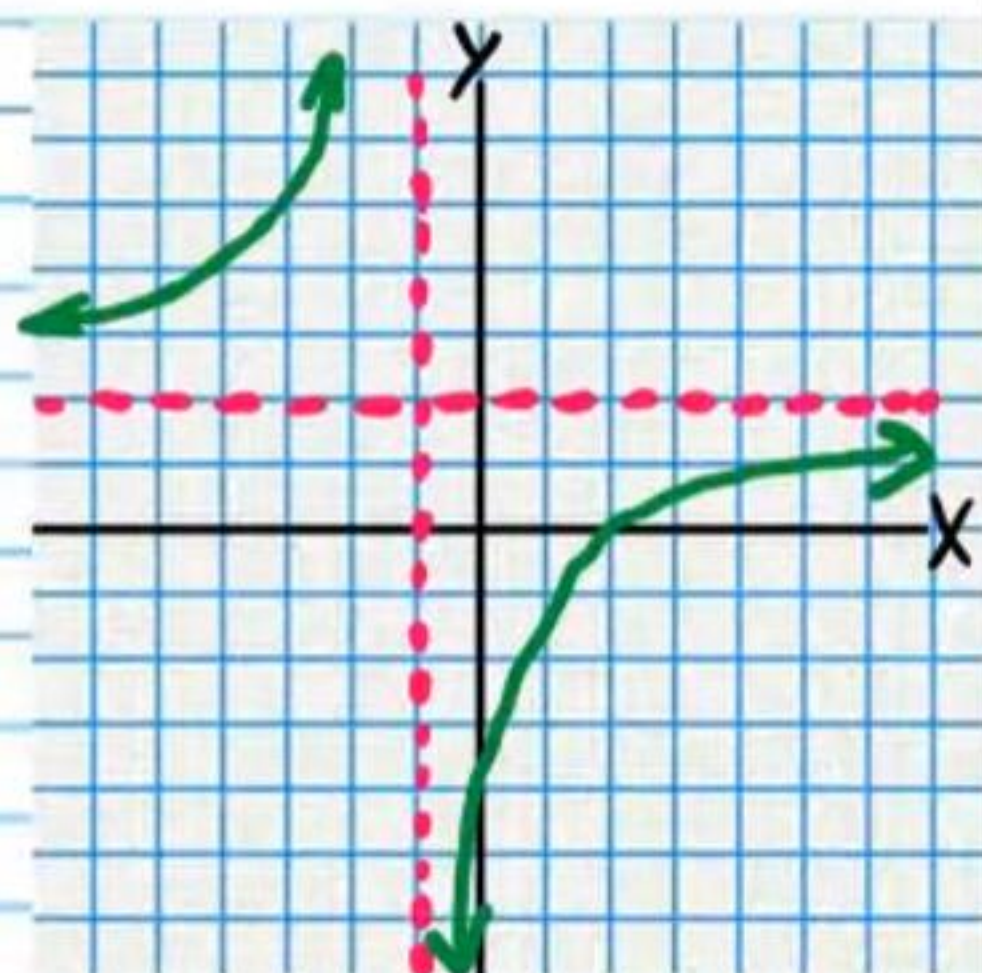
⑮ 

$(-1, 0) \cup (1, 2) \cup (5, \infty)$

①⑥ HA: $y=2$ VA: $x=-1$

Y-Intercept: $(0, -4)$

X-Intercept: $(2, 0)$



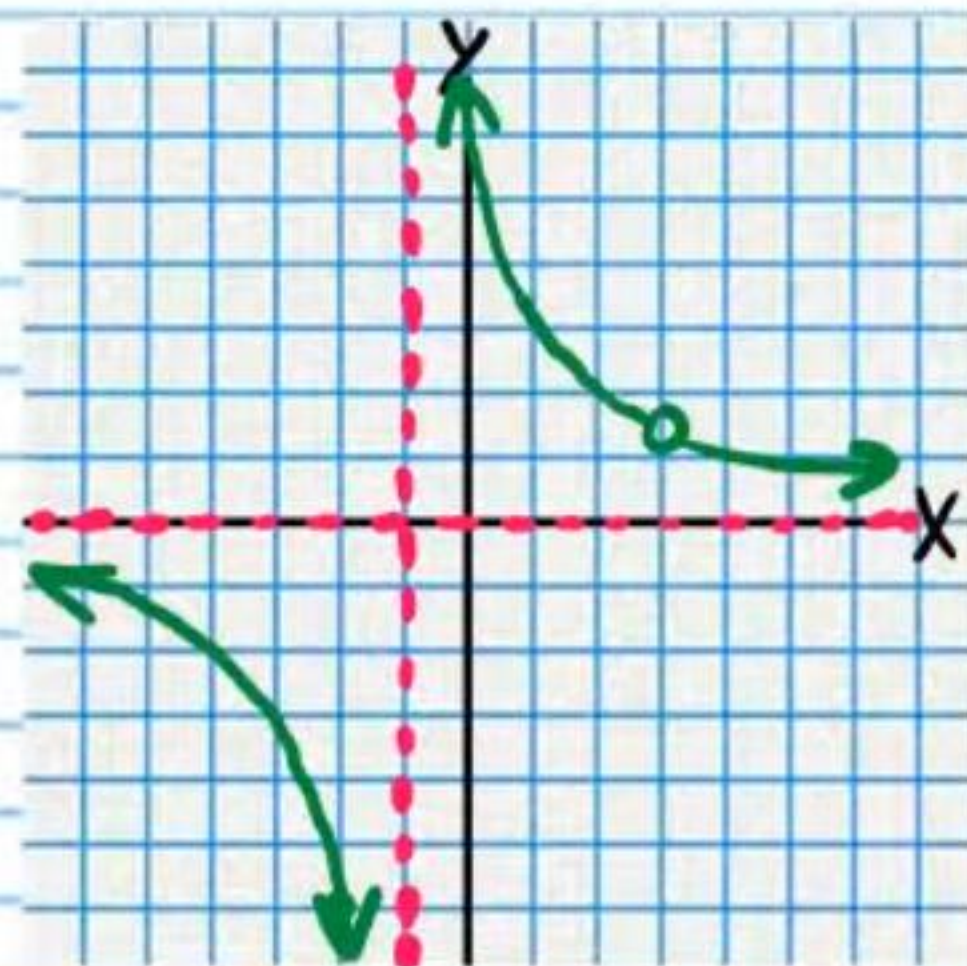
①⑦ Hole: $x=3$

VA: $x=-1$

HA: $y=0$

Y-Intercept: $(0, 6)$

X-Intercept: None

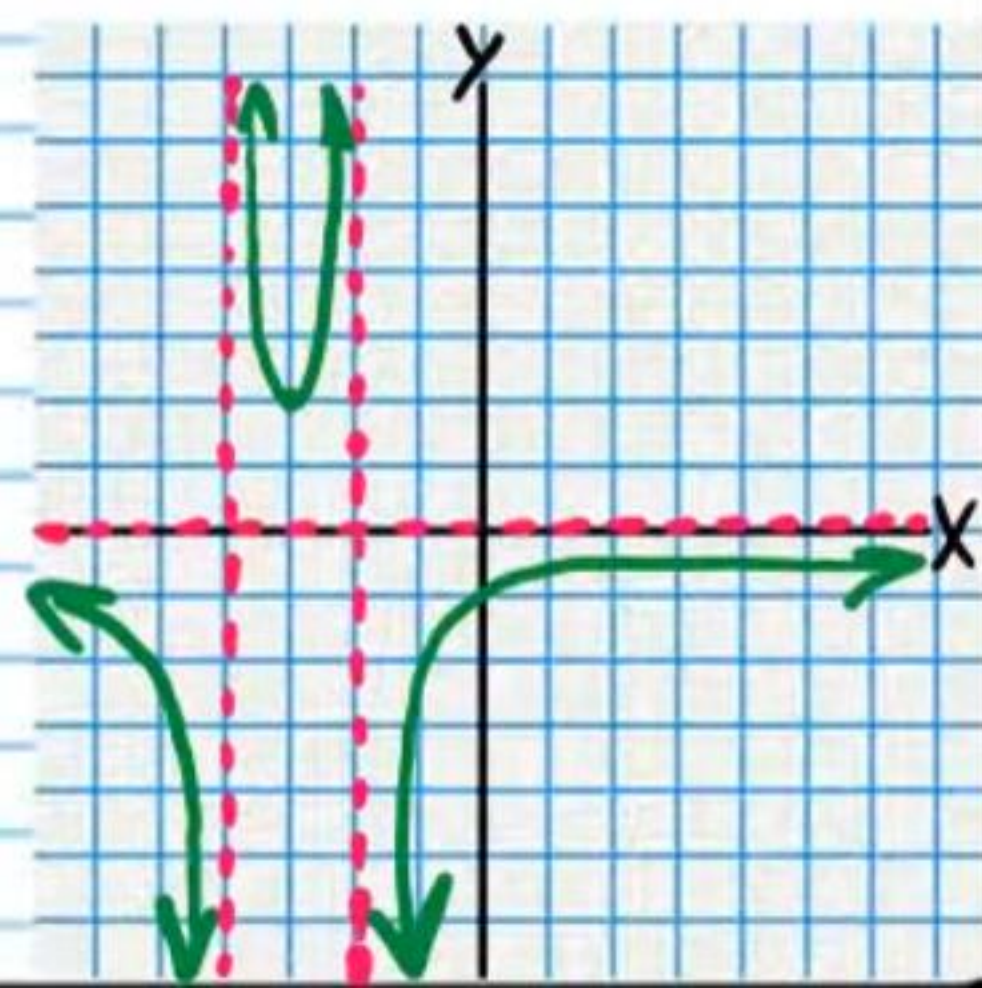


①⑧ VA: $x=-4, x=-2$

HA: $y=0$

Y-Intercept: $(0, -\frac{1}{4})$

X-Intercept: None



①⑨ $x^3 + x^2 + x + 1$

②⑩ $-2x^3 - 4x^2 - 8x - 18 + \frac{35}{-x+2}$

②⑪ $-1, -2, \sqrt{2}, -\sqrt{2}$

②② $-1, \frac{2-i}{3}, \frac{2+i}{3}$

②③ $3 \Rightarrow$ crosses x-axis
 $-1 \Rightarrow$ crosses x-axis
 $f(x) = (x-3)^3(x+1)$

②④ $x = \frac{1}{2}, x = -2, x = 2$

②⑤ Y-Intercept: $(0, 4)$

X-Intercept: $(-4, 0)$

VA: $x = -2$

Asymptote Curve: $x^2 + 2x - 2$